In the present unit we discussing both Markov Chain and Queuing Theory, **Markov Chain:** Introduction to Stochastic Process, Probability Vectors, Stochastic matrices, Regular stochastic matrices, Markov chains, Higher transition probabilities, Stationary distribution of Regular Markov chains and absorbing states, Markov and Poisson processes.
Queuing theory: Introduction, Symbolic representation of a queuing model, Single server Poisson queuing model with infinite capacity (M/M/1 : /FIFO), when and , Performance measures of the model, Single server Poisson queuing model with finite capacity (M/M/S : N/FIFO), Performance measures of the model, Multiple server Poisson queuing model with infinite capacity (M/M/S : /FIFO), when for all , Multiple server Poisson queuing model with finite capacity (M/M/S : N/FIFO), Introduction to M/G/1 queuing model.

Introduction:- A collection of all the outcomes of a random experiment is called as a **sample space** and it is denoted by S. A real variable X is associated with the set of real numbers R for every outcomes of a sample space is called as a **Random Variable** (OR) **Stochastic Variable** (OR) A function mapping from every outcomes of a sample space to the set of a real number is called a random variables. The random variables are represented by $X, Y, Z, ...$ A random variable is depends not only the outcomes of a sample space S and also a parameter t such as a time, then the collection of such a random variables are called as a index set (OR) State space and it is denoted by $X_t(s), \forall s \Subset S$. If such random variable are continuous is called as stochastic process (OR) random process, if it is discrete is called as Markov chain (OR) chain.

Index set: The set used to index the random variable is called index set. The value assumed by the random variable (X) are called states and the set of all possible values form the state space.

A stochastic process can be regarded as a collection of random variables.

Probability Vector: A vector v is said to be a probability vector in which all element are non negative and the sum of all element is equal to unity.

Example - 01: $v = [1, 0]$ is probability vector.

Example - 02: $v = [\frac{1}{2}, \frac{1}{2}]$ is probability vector.

Example - 03: $v_1 = [\frac{1}{3}, \frac{2}{3}, 0]$ is probability vector.

Example - 04: $v_2 = [\frac{1}{2}, \frac{3}{2}, 0]$ is probability vector.
Example - 05: $v_1 = [\frac{-1}{3}, \frac{-2}{3}, 0]$ is not a probability vector.
Stochastic Matrix: A square matrix $P = P_{ij}$ is said to be a stochastic matrix if each of its rows is a probability vector, that is, if each entry of a matrix $P = P_{ij}$ is non negative and the sum of the entries in each row is 1. The stochastic matrix is denoted by $P = P_{ij}$.
Example - 01:

$$P = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} \\ 0 & 1 \end{bmatrix}$$

is a stochastic matrix **Example - 02:**

$$P = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

is a stochastic matrix

Example - 03:

$$P = \begin{bmatrix} 0 & 1 & 0 \\ \frac{1}{2} & 0 & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{4} & \frac{1}{4} \\ \end{bmatrix}$$

stochastic matrix **Example - 04:**

$$P = \begin{bmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & \frac{-1}{3} \\ \end{bmatrix}$$

is not stochastic matrix

. **Regular Stochastic Matrix:** A stochastic matrix $P = P_{ij}$ is said to be a regular stochastic matrix if all the entries of some power P^n are positive.
Example - 01:

$$P = \begin{bmatrix} 0 & 1 \\ \frac{1}{2} & \frac{1}{2} \end{bmatrix}$$

Now consider

$$P^2 = PXP = \begin{bmatrix} 0 & 1 \\ \frac{1}{2} & \frac{1}{2} \end{bmatrix} X \begin{bmatrix} 0 & 1 \\ \frac{1}{2} & \frac{1}{2} \end{bmatrix} = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{4} & \frac{3}{4} \end{bmatrix}$$

Hence the given matrix $P(n = 2)$ is a stochastic matrix.
Properties of a Regular Stochastic Matrix: The following properties are associated with a regular stochastic matrix P of order n
Property - 01: A regular stochastic matrix p has a unique fixed point $x = (x_1, x_2, x_3, ..., x_n)$ such that $xP = x$.
Property - 02: A regular stochastic matrix p has a unique fixed probability vector $v = (v_1, v_2, v_3, ..., v_n)$ such that $vP = v$, where $v_i = \frac{x_i}{\sum_n^{i=1}}$.

Property - 03: A regular stochastic matrix p then $P^2, P^3, \ldots$ approaches the matrix V whose rows are each in the fixed probability vector v.
Property - 04: If u be any probability vector and P be a regular stochastic matrix then the sequence of vectors $uP, uP^2, uP^3, \ldots$ approaches the unique fixed probability vector v.
Markov Chains: We now consider a sequence of trials whose outcomes, say, $(X_1, X_2, X_3, \ldots)$ satisfy the following two properties.
Property - 05 Each outcome belong to a finite set of outcomes $(a_1, a_2, a_3, \ldots, a_m)$ called the state space of the system, if the outcome on the n^{th} trial is a_i, then we say that the system is in state a_i at time n or at the n^{th} step.
Property - 06 The outcome of any trial depends at most upon the outcome of the immediately preceding trial and not upon any other previous outcome, with each pair of states (a_i, a_j) there is given the probability P_{ij} occurs immediately after a_i occurs. Such a stochastic process is called a (finite) Markov chain. The element of the matrix P_{ij} called the transition probabilities
Fixed Probability Vector: Given a regular stochastic matrix P of order m, if there exists a probability vector v of order m such that $vP = v$, then v is called a fixed probability vector of a regular stochastic matrix P. It can be proved that a vector v exists and is unique.

Problem - 01: Find the fixed probability vector of the regular stochastic matrix

$$P = \begin{bmatrix} \frac{3}{4} & \frac{1}{4} \\ \frac{1}{2} & \frac{1}{2} \end{bmatrix}$$

Solution: Given a regular stochastic matrix

$$P = \begin{bmatrix} \frac{3}{4} & \frac{1}{4} \\ \frac{1}{2} & \frac{1}{2} \end{bmatrix}$$

We have, by the property of the regular stochastic matrix, to find the fixed probability vector $v = \begin{bmatrix} x, & y \end{bmatrix}$, where $x + y = 1$ –(1) such that $vP = v$

$$vP = \begin{bmatrix} x, & y \end{bmatrix} \begin{bmatrix} \frac{3}{4} & \frac{1}{4} \\ \frac{1}{2} & \frac{1}{2} \end{bmatrix} = \begin{bmatrix} x, & y \end{bmatrix}$$

$$vP = \begin{bmatrix} \frac{3x}{4} + \frac{y}{2}, & \frac{x}{4} + \frac{y}{2} \end{bmatrix} = \begin{bmatrix} x, & y \end{bmatrix}$$

. comparing on both sides $\frac{3x}{4} + \frac{y}{2} = x$ and $\frac{x}{4} + \frac{y}{2} = y$, from equation (1), $y = 1 - x$
$\frac{3x}{4} + \frac{(1-x)}{2} = x$, $\frac{3x+2-2x}{4} = x$, $\frac{x+2}{4} = x$, $x + 2 = 4x$, $3x = 2$, $x = \frac{2}{3}$ and
$y = 1 - x = 1 - \frac{2}{3}$, $y = \frac{1}{3}$

Hence the fixed probability vector is $v = \begin{bmatrix} x, y \end{bmatrix} = \begin{bmatrix} \frac{2}{3}, \frac{1}{3} \end{bmatrix}$

Problem - 02: Find the fixed probability vector of the regular stochastic matrix

$$A = \begin{bmatrix} 0 & 1 & 0 \\ \frac{1}{6} & \frac{1}{2} & \frac{1}{3} \\ 0 & \frac{2}{3} & \frac{1}{3} \end{bmatrix}$$

Solution: Given a regular stochastic matrix

$$A = \begin{bmatrix} 0 & 1 & 0 \\ \frac{1}{6} & \frac{1}{2} & \frac{1}{3} \\ 0 & \frac{2}{3} & \frac{1}{3} \end{bmatrix}$$

We have, by the property of the regular stochastic matrix, to find the fixed probability vector $v = \begin{bmatrix} x & y & z \end{bmatrix}$, where $x + y + z = 1$ –(1) such that $vA = v$

$$vA = \begin{bmatrix} x, & y, & z \end{bmatrix} \begin{bmatrix} 0 & 1 & 0 \\ \frac{1}{6} & \frac{1}{2} & \frac{1}{3} \\ 0 & \frac{2}{3} & \frac{1}{3} \end{bmatrix} = \begin{bmatrix} x, & y, & z \end{bmatrix}$$

$$\begin{bmatrix} \frac{y}{6}, & x + \frac{y}{2} + \frac{2z}{3}, & \frac{y}{3} + \frac{z}{3} \end{bmatrix} = \begin{bmatrix} x, & y, & z \end{bmatrix}$$

comparing on both sides

$$\frac{y}{6} = x, x + \frac{y}{2} + \frac{2z}{3} = y, \frac{y}{3} + \frac{z}{3} = z$$

$$y = 6x, 6x + 3y + 4z = 6y, y + z = 3z$$

$$y = 6x, 6x - 3y + 4z = 0, y - 2z = 0$$

From the equation (1) $z = 1 - x - y$, using $y = 6x$,$z = 1 - x - 6x$, $z = 1 - 7x$ using $y = 6x$ and $z = 1 - 7x$ in $6x - 18x + 4 - 28x = 0$, $x = \frac{1}{10}$, $y = \frac{6}{10}$ and $z = \frac{3}{10}$. Hence the fixed probability vector is $v = \begin{bmatrix} x, & y, & z \end{bmatrix} = \begin{bmatrix} \frac{1}{10}, & \frac{6}{10}, & \frac{3}{10} \end{bmatrix}$.

Markov Chain: a markov chain is a stochastic process with discrete states (finite or countable states) operating in discrete time in which the probability of moving from one state to another are dependent only on the present state of the process.
Transition Probability: The probability of moving from one state to another state or remaining in the same state during a single time period is called the transition probability. The matrix associated with the transition probability is called transition probability matrix(TPM) and it is denoted by $P - P_{ij}$, where

$$P_{ij} = \begin{bmatrix} p_{11} & p_{12} & p_{13} \\ p_{21} & p_{22} & p_{23} \\ p_{31} & p_{32} & p_{33} \\ & & \end{bmatrix}_{3X3}$$

The element of a TPM P have the following properties

1. $0 \leq P_{ij} \leq 1$.
2. $\sum P_{ij} = 1$ (Row sum is equal to 1)

Example - 01: Three boys A, B, C are throwing a ball to each other. A always throws the ball to B and B always throws the ball to C. C is just as likely to throw the ball to B as to A.

Solution: Therefore, the state space for this example is

State Space $= \left\{ A, B, C \right\}$

The corresponding transition probability matrix (TPM) P_{ij} is

$$P_{ij} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ \frac{1}{2} & \frac{1}{2} & 0 \\ \end{bmatrix}_{3X3}$$

Example - 02: A person commutes the distance to his office every day either by train or by bus. Suppose he does not go by train for two consecutive days, but if he goes by bus the next day he is just as likely to go by bus again as he is to travel by train.

Solution: Therefore, the state space for this example is

State Space $= \left\{ Train(T), Bus(B) \right\}$

The corresponding transition probability matrix (TPM) P_{ij} is

$$P_{ij} = \begin{bmatrix} 0 & 1 \\ \frac{1}{2} & \frac{1}{2} \\ \end{bmatrix}_{2X2}$$

Higher Transition Probabilities: The probability that the system changes from one state to another state in exactly n steps and it is denoted by P_{ij}^{n}. The matrix formed by the probability P_{ij}^{n} is called the $n-$ step transition matrix and it is denoted by P^n.

Example: Three boys A, B, C are throwing a ball to each other. A always throws the ball to B and B always throws the ball to C. C is just as likely to throw the ball to B as to A.

Solution: Therefore, the state space for this example is

State Space $= \left\{ A, B, C \right\}$

The corresponding transition probability matrix (TPM) P_{ij} is

$$P = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ \frac{1}{2} & \frac{1}{2} & 0 \end{bmatrix}_{3X3}$$

Now consider

$$P^2 = P.P = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ \frac{1}{2} & \frac{1}{2} & 0 \end{bmatrix} X \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ \frac{1}{2} & \frac{1}{2} & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 1 \\ \frac{1}{2} & \frac{1}{2} & 0 \\ 0 & \frac{1}{2} & \frac{1}{2} \end{bmatrix}$$

$$P^3 = P.P^2 = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ \frac{1}{2} & \frac{1}{2} & 0 \end{bmatrix} X \begin{bmatrix} 0 & 0 & 1 \\ \frac{1}{2} & \frac{1}{2} & 0 \\ 0 & \frac{1}{2} & \frac{1}{2} \end{bmatrix} = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} & 0 \\ 0 & \frac{1}{2} & \frac{1}{2} \\ \frac{1}{4} & \frac{1}{4} & \frac{1}{2} \end{bmatrix}$$

$$P^4 = P.P^3 = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ \frac{1}{2} & \frac{1}{2} & 0 \end{bmatrix} X \begin{bmatrix} \frac{1}{2} & \frac{1}{2} & 0 \\ 0 & \frac{1}{2} & \frac{1}{2} \\ \frac{1}{4} & \frac{1}{4} & \frac{1}{2} \end{bmatrix} = \begin{bmatrix} 0 & \frac{1}{2} & \frac{1}{2} \\ \frac{1}{4} & \frac{1}{4} & \frac{1}{2} \\ \frac{1}{4} & \frac{1}{2} & \frac{1}{4} \end{bmatrix}$$

$$P^5 = P.P^4 = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ \frac{1}{2} & \frac{1}{2} & 0 \end{bmatrix} X \begin{bmatrix} 0 & \frac{1}{2} & \frac{1}{2} \\ \frac{1}{4} & \frac{1}{4} & \frac{1}{2} \\ \frac{1}{4} & \frac{1}{2} & \frac{1}{4} \end{bmatrix} = \begin{bmatrix} \frac{1}{4} & \frac{1}{4} & \frac{1}{2} \\ \frac{1}{4} & \frac{1}{2} & \frac{1}{4} \\ \frac{1}{8} & \frac{3}{8} & \frac{1}{2} \end{bmatrix}$$

Let us assume that C was the person having the ball first then the initial probability vector is $P^{(0)} = \begin{bmatrix} 0 & 0 & 1 \end{bmatrix}$ and consider

$$P^5 = P^{(0)}.P^5 = \begin{bmatrix} 0 & 0 & 1 \end{bmatrix} X \begin{bmatrix} \frac{1}{4} & \frac{1}{4} & \frac{1}{2} \\ \frac{1}{4} & \frac{1}{2} & \frac{1}{4} \\ \frac{1}{8} & \frac{3}{8} & \frac{1}{2} \end{bmatrix} = \begin{bmatrix} \frac{1}{8} & \frac{3}{8} & \frac{1}{2} \end{bmatrix}$$

$$P^5 = \begin{bmatrix} P_A^5, & P_B^5, & P_C^5 \end{bmatrix} = \begin{bmatrix} \frac{1}{8}, & \frac{3}{8}, & \frac{1}{2} \end{bmatrix}$$

Hence, this implies that after 5th throws the probability that the ball is with a boy A is $\frac{1}{8}$, the ball with a boy B is $\frac{3}{8}$ and the ball with a boy C is $\frac{1}{2}$.

Stationary Probability of Regular Markov Chain:

A markov chain is said to be regular if the associated transition probability matrix P is regular stochastic matrix.

If P is a regular stochastic matrix of the Markov chain, then the sequences of n step transition matrix $P^2, P^3, P^4, ..., P^n$ approaches the matrix v whose rows are each the unique fixed probability vector v of a matrix P.

$$P^n = P^{(0)}.P^n$$

where

$$P^n = \begin{bmatrix} P_1^n, & P_2^n, & P_3^n, ..., P_m^n \end{bmatrix}$$

Irreducible:A Markov chain is said to be irreducible if every state can be reached from every other state in a finite number of steps.

(OR)

A Markov chain is said to be irreducible if the associated transition probability matrix p is regular.

Classification of a Markov Chain:

Absorbing State: A state in a Markov chain is said to be an absorbing state if once the state is entered it continuous to remain in the same state.
In Markov chain is in an absorbing state then the transition probabilities P_{ij} are such that

$$P^{ij} = 1, fori = jandP_{ij} = 0, otherwise$$

Thus a state a_i of the Markov chain is in absorbing state if the i^{th} row of the transition probability matrix P has 1 on the principal diagonal and zeros elsewhere.

Examples: Consider the Markov chain having the following matrix as its transition probability matrix

Example - 01:

$$P = \begin{bmatrix} 1 & 0 & 0 \\ 0.3 & 0.7 & 0 \\ 0 & 0.2 & 0.8 \\ \end{bmatrix}$$

Here, the state A is an absorbing state and states B and C are not in an absorbing state.

Example - 02:

$$P = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$$

Here, the state B is an absorbing state and states A and C are not in an absorbing state.

Example - 03:

$$P = \begin{bmatrix} \frac{1}{4} & \frac{1}{4} & \frac{1}{4} & \frac{1}{4} \\ \frac{1}{2} & 0 & 0 & \frac{1}{4} \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Here, the states A and B are not an absorbing state and states C and D are in an absorbing state.

Recurrent State and Transition State: A state in a Markov chain is said to be Recurrent state if starting from that state it returns to the same state again, otherwise, the state is said to be a transition state.

Examples: consider the following Markov chain having the transition probability matrix P is

$$P = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

$$P^2 = P.P = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} X \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$P^3 = P.P^2 = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} X \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

Since $P^3 = P$, the state return to the same state after two steps. Hence the state is in recurrent state.

Problems:

Problem - 01: The transition probability matrix of a Markov chain is

$$P = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{3}{4} & \frac{1}{4} \end{bmatrix}$$

with the initial probability distribution $P^{(0)} = \begin{bmatrix} \frac{1}{4} & \frac{3}{4} \end{bmatrix}$. Find (i) $P_{21}^{(2)}, P_{12}^{(2)}, P^2, P_1^{(2)}, P_2^{(2)}$, (ii) the vector $P^{(0)}P^n$ approaches and (iii) the matrix P^n approaches.

Solution: Given (i)

$$P = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{3}{4} & \frac{1}{4} \end{bmatrix}$$

and $P^{(0)} = \begin{bmatrix} \frac{1}{4} & \frac{3}{4} \end{bmatrix}$

$$P^2 = P.P = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{3}{4} & \frac{1}{4} \end{bmatrix} X \begin{bmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{3}{4} & \frac{1}{4} \end{bmatrix} = \begin{bmatrix} \frac{5}{8} & \frac{3}{8} \\ \frac{9}{16} & \frac{7}{16} \end{bmatrix} = \begin{bmatrix} P_{11}^{(2)} & P_{12}^{(2)} \\ P_{21}^{(2)} & P_{22}^{(2)} \end{bmatrix}$$

$\therefore P_{21}^{(2)} = \frac{9}{16}$ and $P_{12}^{(2)} = \frac{3}{8}$
We know that, $P^{(n)} = P^{(0)}P^{(n)}$

$$P^2 = P^{()0}.P^2 = \begin{bmatrix} \frac{1}{4} & \frac{3}{4} \end{bmatrix} X \begin{bmatrix} \frac{5}{8} & \frac{3}{8} \\ \frac{9}{16} & \frac{7}{16} \end{bmatrix} = \begin{bmatrix} \frac{37}{64} & \frac{27}{64} \end{bmatrix} = \begin{bmatrix} P_1^{(2)} & P_2^{(2)} \end{bmatrix}$$

$\therefore P_1^{(2)} = \frac{37}{64}$ and $P_2^{(2)} = \frac{27}{64}$
(ii) The vector $P^{(0)}P^n$ approaches the unique fixed probability vector of P. we have, by the property of TPM, let $v = \begin{bmatrix} x & y \end{bmatrix}$, where $x + y = 1$ such that $vP = v$.
Now consider $vP = v$

$$\begin{bmatrix} x & y \end{bmatrix} X \begin{bmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{3}{4} & \frac{1}{4} \end{bmatrix} = \begin{bmatrix} x & y \end{bmatrix}$$

$$\begin{bmatrix} \frac{x}{2} + \frac{3y}{4} & \frac{x}{2} + \frac{y}{4} \end{bmatrix} = \begin{bmatrix} x & y \end{bmatrix}$$

Equating on both sides $\frac{x}{2} + \frac{3y}{4} = x$ and $\frac{x}{2} + \frac{y}{4} = y$, by solving these equations with the equation $y = 1 - x$, we get $x = \frac{3}{5}$ and $\frac{2}{5}$.

Hence the vector $P^{(0)}P^n$ approaches the vector $v = \begin{bmatrix} \frac{3}{5} & \frac{2}{5} \end{bmatrix}$

(iii) P^n approaches the vector v whose rows are each of the fixed probability vector of the matrix P. Therefore, the matrix P^n approaches the matrix

$$p^n = \begin{bmatrix} \frac{3}{5} & \frac{2}{5} \\ \frac{3}{5} & \frac{2}{5} \end{bmatrix}$$

Problem - 02: The transition probability matrix of a Markov chain is given by

$$P = \begin{bmatrix} \frac{1}{2} & 0 & \frac{1}{2} \\ 1 & 0 & 0 \\ \frac{1}{4} & \frac{1}{2} & \frac{1}{4} \end{bmatrix}$$

and the initial probability distribution is $P^{(0)} = \begin{bmatrix} \frac{1}{2}, & \frac{1}{2}, & 0 \end{bmatrix}$. Find $P^2, P_{13}^{(2)}, P_{23}^{(2)}, P_{32}^{(2)}, P_{33}^{(2)}, P_1^{(2)}, P_2^{(2)}$.

Solution: Given

$$P = \begin{bmatrix} \frac{1}{2} & 0 & \frac{1}{2} \\ 1 & 0 & 0 \\ \frac{1}{4} & \frac{1}{2} & \frac{1}{4} \end{bmatrix}$$

and the initial probability distribution is $P^{(0)} = \begin{bmatrix} \frac{1}{2}, & \frac{1}{2}, & 0 \end{bmatrix}$ Now, first find

$$P^2 = P.P = \begin{bmatrix} \frac{1}{2} & 0 & \frac{1}{2} \\ 1 & 0 & 0 \\ \frac{1}{4} & \frac{1}{2} & \frac{1}{4} \end{bmatrix} X \begin{bmatrix} \frac{1}{2} & 0 & \frac{1}{2} \\ 1 & 0 & 0 \\ \frac{1}{4} & \frac{1}{2} & \frac{1}{4} \end{bmatrix} = \begin{bmatrix} \frac{3}{8} & \frac{1}{4} & \frac{3}{8} \\ \frac{1}{2} & 0 & \frac{1}{2} \\ \frac{1}{4} & \frac{1}{2} & \frac{1}{4} \end{bmatrix}$$

Hence, From the above matrix $P_{13}^{(2)} = \frac{3}{8}$, $P_{23}^{(2)} = \frac{1}{2}$, $P_{23}^{(2)} = \frac{1}{2}$ and $P_{33}^{(2)} = \frac{1}{4}$.
We have, $P^{(n)} = P^{(0)}.P^{(n)}$, $P^{(2)} = P^{(0)}.P^{(2)}$

Continued Unit - III Problems on Markov chain ...

$$P^{(2)} = P^{(0)}.P^{(2)} = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} & 0 \end{bmatrix} X \begin{bmatrix} \frac{3}{8} & \frac{1}{4} & \frac{3}{8} \\ \frac{1}{2} & 0 & \frac{1}{2} \\ \frac{1}{4} & \frac{1}{2} & \frac{1}{4} \end{bmatrix} = \begin{bmatrix} \frac{7}{16}, & \frac{1}{8}, & \frac{7}{16} \end{bmatrix} = \Big[\quad P_1^{(2)}, \quad P_2^{(}$$

$$P^{(2)} = P^{(0)}.P^{(2)} = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} & 0 \end{bmatrix} X \begin{bmatrix} \frac{3}{8} & \frac{1}{4} & \frac{3}{8} \\ \frac{1}{2} & 0 & \frac{1}{2} \\ \frac{1}{4} & \frac{1}{2} & \frac{1}{4} \end{bmatrix} = \begin{bmatrix} \frac{7}{16}, & \frac{1}{8}, & \frac{7}{16} \end{bmatrix} = \Big[\quad P_1^{(2)}, \quad P_2^{(}$$

Therefore, $P_1^{(2)} = \frac{7}{16}$, $P_2^{(2)} = \frac{1}{8}$ and $P_3^{(2)} = \frac{7}{16}$.
Problem - 03: Prove that the Markov chain with the transition probability matrix

$$P = \begin{bmatrix} 0 & \frac{2}{3} & \frac{1}{3} \\ \frac{1}{2} & 0 & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} & 0 \end{bmatrix}$$

is an Irreducible. Find the corresponding stationary probability vector.
Solution: Given the TPM

$$P = \begin{bmatrix} 0 & \frac{2}{3} & \frac{1}{3} \\ \frac{1}{2} & 0 & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} & 0 \end{bmatrix}$$

To prove that the Markov chain is Irreducible we need to show that the TPM P ia d regular stochastic matrix

$$P^2 = P.P = \begin{bmatrix} 0 & \frac{2}{3} & \frac{1}{3} \\ \frac{1}{2} & 0 & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} & 0 \end{bmatrix} X \begin{bmatrix} 0 & \frac{2}{3} & \frac{1}{3} \\ \frac{1}{2} & 0 & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} & 0 \end{bmatrix} = \begin{bmatrix} \frac{1}{2} & \frac{1}{6} & \frac{1}{3} \\ \frac{1}{4} & \frac{7}{12} & \frac{1}{6} \\ \frac{1}{4} & \frac{1}{3} & \frac{5}{12} \end{bmatrix}$$

Since all the entries in P^2 are positive we conclude that the TPM P is a regular stochastic matrix. Hence the Markov chain having the TPM P is an Irreducible.

To find the stationary probability vector:

Let $v = \begin{bmatrix} x & y & Z \end{bmatrix}$, where $x + y + z = 1$ such that $vP = v$.

Now consider $vP = v$

$$vXP = \begin{bmatrix} x & y & z \end{bmatrix} X \begin{bmatrix} 0 & \frac{2}{3} & \frac{1}{3} \\ \frac{1}{2} & 0 & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} & 0 \end{bmatrix} = \begin{bmatrix} x & y & Z \end{bmatrix}$$

$$\begin{bmatrix} \frac{y}{2} + \frac{z}{2}, & \frac{2x}{3} + \frac{z}{2}, & \frac{x}{2} + \frac{y}{2} \end{bmatrix} = \begin{bmatrix} x & y & Z \end{bmatrix}$$

Equating on both sides, we get $\frac{y}{2} + \frac{z}{2} = x$, $\frac{2x}{3} + \frac{z}{2} = y$ and $\frac{x}{2} + \frac{y}{2} = z$, by solving these equations with the equation $x + y + z = 1$, we get $x = \frac{1}{3}$, $y = \frac{10}{27}$ and $z = \frac{8}{27}$.

Therefore $v = \begin{bmatrix} \frac{1}{3}, & \frac{10}{27}, & \frac{8}{27} \end{bmatrix}$ is the required stationary probability vector.

Problem - 04: A Habitual gambler is a member of two clubs A and B. He visits either of the clubs everyday for playing cards. He never visits club A on two consecutive days. But if he visits club B on a particular day, then the next day he is as likely to visit club B or club A. Find the transition probability matrix P of the Markov chain. Also, (i) Show that the transition probability matrix P is a regular stochastic matrix and find the unique fixed probability vector. (ii) If the person had visited club B on Monday, find the probability that he visits club A on Thursday.

Solution: Here, there are two possible states, $A =$ represents he visit club A and $B =$ represents he visit club B and the state space is

State space $= \begin{bmatrix} A, & B \\ & \end{bmatrix}$ and the corresponding the transition probability matrix P is

$$P = \begin{bmatrix} 0 & 1 \\ \frac{1}{2} & \frac{1}{2} \\ & \end{bmatrix}$$

(i) To show that the TPM P is regular stochastic matrix we need to find P^2

$$P^{(2)} = P.P = \begin{bmatrix} 0 & 1 \\ \frac{1}{2} & \frac{1}{2} \\ & \end{bmatrix} X \begin{bmatrix} 0 & 1 \\ \frac{1}{2} & \frac{1}{2} \\ & \end{bmatrix} = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{4} & \frac{3}{4} \\ & \end{bmatrix}$$

Since all the entries in $P^{(2)}$ are positive then we conclude that the TPM P is a regular stochastic matrix.

To find the fixed probability vector:

Let $v = \begin{bmatrix} x & y \end{bmatrix}$, where $x + y = 1$ such that $vP = v$.

Now consider $vP = v$

$$vXP = \begin{bmatrix} x & y \end{bmatrix} X \begin{bmatrix} 0 & 1 \\ \frac{1}{2} & \frac{1}{2} \end{bmatrix} = \begin{bmatrix} x & y \end{bmatrix}$$

$$\begin{bmatrix} \frac{y}{2} & x + \frac{y}{2} \end{bmatrix} = \begin{bmatrix} x & y \end{bmatrix}$$

By equating on both sides, $\frac{y}{2} = x$ and $x + \frac{y}{2} = y$, solving these equations with the equation $y = 1 - x$, we get $x = \frac{1}{3}$ and $y = \frac{2}{3}$. Hence the fixed probability vector is $v = \begin{bmatrix} \frac{1}{3} & \frac{2}{3} \end{bmatrix}$.

(ii) consider Monday to Tuesday as day - 1, Tuesday to Wednesday as day - 2 and Wednesday to Thursday as day - 3, total three days, then the probability that he visits club A after three days is equivalent to finding $P_{21}^{(3)}$ from P^3.

$$P^{(3)} = P^{(2)}.P = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{4} & \frac{3}{4} \end{bmatrix}$$

$$\text{X} \begin{bmatrix} 0 & 1 \\ \frac{1}{4} & \frac{3}{4} \end{bmatrix} = \begin{bmatrix} \frac{1}{4} & \frac{3}{4} \\ \frac{3}{8} & \frac{5}{8} \end{bmatrix}$$

$$P^{(3)} = \begin{bmatrix} \frac{1}{4} & \frac{3}{4} \\ \frac{3}{8} & \frac{5}{8} \end{bmatrix} = \begin{bmatrix} P_{11}^{(3)} & P_{12}^{(3)} \\ P_{21}^{(3)} & P_{22}^{(3)} \end{bmatrix}$$

Therefore, The probability that he visits a club A on Thursday is $P_{21}^{(3)} = \frac{3}{8}$.

Problem - 05: A student study habits are as follows. If he studies one night, he is 70% sure not to study the next night. On the other hand if he does not study one night, he is 60% sure not to study the next night. In the long run how often does he study?

Solution: Here, there are two possible study habits are $A =$ represents the studying and $B =$ represents the not studying then the state space $= \begin{bmatrix} A & B \end{bmatrix}$

The associated transition probability matrix P is

$$P = \begin{bmatrix} \frac{30}{100} & \frac{70}{100} \\ \frac{40}{100} & \frac{60}{100} \end{bmatrix} = \begin{bmatrix} 0.3 & 0.7 \\ 0.6 & 0.6 \end{bmatrix}$$

$$\begin{bmatrix} 0.3x + 0.4y, & 0.7x + 0.6y \end{bmatrix} = \begin{bmatrix} x & y \end{bmatrix}$$

Equating on both sides, we get $0.3x + 0.4y = x$ and $0.7x + 0.6y = y$, by solving these equations $x = \frac{4}{11}$ and $y = \frac{7}{11}$

$v = \begin{bmatrix} x & y \end{bmatrix} = \begin{bmatrix} \frac{4}{11} & \frac{7}{11} \end{bmatrix} = \begin{bmatrix} P_A & P_B \end{bmatrix}$ Hence we conclude that in the long run the student will study is $\frac{4}{11}$ of the time or 36.3636% of the time.

Problem - 06: Three boys A, B and C are throwing a ball to each other. A always throws the ball to B and B always throws the ball to C. But C is just as likely to throw the ball to B as to A. If C was the first person to throw the ball, find the probabilities that (i) A has the ball, (ii) B has the ball and (iii) C has the ball, for the fourth throw.

Solution: Here, there are three possible states, (a_1, a_2) and a_3, a_1 = denotes the a boy A has a ball, a_2 = denotes the a boy B has a ball and a_3 = denotes the a boy C has a ball.

The state space $= \begin{bmatrix} a_1 & a_2 & a_3 \end{bmatrix}$. The transition probability matrix P is

$$P = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ \frac{1}{2} & \frac{1}{2} & 0 \end{bmatrix}$$

$P_{11} = P\left(a_1/a_1\right) = 0 =$ From the state a_1 to the state $a_1 =$ The probability that the ball is thrown from A to A.
$P_{12} = P\left(a_1/a_2\right) = 1 =$ From the state a_1 to the state $a_2 =$ The probability that the ball is thrown from A to B.
$P_{13} = P\left(a_1/a_3\right) = 0 =$ From the state a_1 to the state $a_3 =$ The probability that the ball is thrown from A to C.
$P_{21} = P\left(a_2/a_1\right) = 0 =$ From the state a_2 to the state $a_1 =$ The probability that the ball is thrown from B to A.
$P_{22} = P\left(a_2/a_2\right) = 0 =$ From the state a_2 to the state $a_2 =$ The probability that the ball is thrown from B to B.
$P_{23} = P\left(a_2/a_3\right) = 1 =$ From the state a_2 to the state $a_3 =$ The probability that the ball is thrown from B to C.

$P_{31} = P\Big(a_3/a_1\Big) = \frac{1}{2}$ = From the state a_3 to the state a_1 = The probability that the ball is thrown from C to A.
$P_{32} = P\Big(a_3/a_2\Big) = \frac{1}{2}$ = From the state a_3 to the state a_2 = The probability that the ball is thrown from C to B.
$P_{33} = P\Big(a_3/a_3\Big) = 0$ = From the state a_3 to the state a_3 = The probability that the ball is thrown from C to C.
Now, if the ball was first thrown by C, then the initial probability vector is

$$P^{(0)} = \begin{bmatrix} P_A^{(0)} & P_B^{(0)} & P_C^{(0)} \end{bmatrix} = \begin{bmatrix} 0 & 0 & 1 \end{bmatrix}.$$

After fourth throws, the probability distributions is given by $P^4 = P^{(0)} X P^{(4)}$

$$P^{(3)} = P^{(0)} X P^{(3)} = \begin{bmatrix} 0 & 0 & 1 \end{bmatrix} X \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ \frac{1}{2} & \frac{1}{2} & 0 \end{bmatrix} X \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ \frac{1}{2} & \frac{1}{2} & 0 \end{bmatrix} X \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ \frac{1}{2} & \frac{1}{2} & 0 \end{bmatrix}$$

$$P^{(4)} = P^{(0)} X P^{(3)} X P = \begin{bmatrix} 0 & 0 & 1 \end{bmatrix} X \begin{bmatrix} \frac{1}{2} & \frac{1}{2} & 0 \\ 0 & \frac{1}{2} & \frac{1}{2} \\ \frac{1}{4} & \frac{1}{4} & \frac{1}{2} \end{bmatrix} X \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ \frac{1}{2} & \frac{1}{2} & 0 \end{bmatrix}$$

$$P^{(4)} = P^{(0)} X P^{(4)} = \begin{bmatrix} 0 & 0 & 1 \end{bmatrix} X \begin{bmatrix} 0 & \frac{1}{2} & \frac{1}{2} \\ \frac{1}{4} & \frac{1}{4} & \frac{1}{2} \\ \frac{1}{4} & \frac{1}{2} & \frac{1}{4} \end{bmatrix}$$

$$P^{(4)} = \begin{bmatrix} P_A^{(4)} & P_B^{(4)} & P_C^{(4)} \end{bmatrix} = \begin{bmatrix} \frac{1}{4} & \frac{1}{2} & \frac{1}{4} \end{bmatrix}$$

Hence, after fourth throws, the probability that the ball is with (i) A is $P_A^{(4)} = \frac{1}{4}$, (ii) B is $P_B^{(4)} = \frac{1}{2}$ and (iii) C is $P_C^{(4)} = \frac{1}{4}$.

Problem - 07: Consider repeated tosses of a fair die. Let Xn be the maximum of the numbers occurring in the first n trials. (i) Find transition matrix of the markov chain (ii) Is this matrix regular(iii) Find the probability distribution after first toss (iv) Find $P^{(2)}$ and $P^{(3)}$.

Solution: Here, there are six possible states and the state space $= \begin{bmatrix} 1, & 2, & 3, & 4, & 5, & 6 \end{bmatrix}$. Let $X_n =$ the maximum of the number occurring in the first n trials $= 3$, say then $X_{n+1} = 3$, if the $(n+1)^{th}$ trial results in $1, 2$ or 3
$X_{n+1} = 4$, if the $(n+1)^{th}$ trial results in 4
$X_{n+1} = 5$, if the $(n+1)^{th}$ trial results in 5
$X_{n+1} = 6$, if the $(n+1)^{th}$ trial results in 6

$$P(X_{n+1} = 3/X_n = 3) = \frac{1}{6} + \frac{1}{6} + \frac{1}{6} = \frac{3}{6}$$

$$P(X_{n+1} = i/X_n = 3) = \frac{1}{6}, wheni = 4, 5, 6$$

The transition probability matrix P is

$$P = \begin{bmatrix} \frac{1}{6} & \frac{1}{6} & \frac{1}{6} & \frac{1}{6} & \frac{1}{6} & \frac{1}{6} \\ 0 & \frac{2}{6} & \frac{1}{6} & \frac{1}{6} & \frac{1}{6} & \frac{1}{6} \\ 0 & 0 & \frac{3}{6} & \frac{1}{6} & \frac{1}{6} & \frac{1}{6} \\ 0 & 0 & 0 & \frac{4}{6} & \frac{1}{6} & \frac{1}{6} \\ 0 & 0 & 0 & 0 & \frac{5}{6} & \frac{1}{6} \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

$$P^{(2)} = PXP = \begin{bmatrix} \frac{1}{6} & \frac{1}{6} & \frac{1}{6} & \frac{1}{6} & \frac{1}{6} & \frac{1}{6} \\ 0 & \frac{2}{6} & \frac{1}{6} & \frac{1}{6} & \frac{1}{6} & \frac{1}{6} \\ 0 & 0 & \frac{3}{6} & \frac{1}{6} & \frac{1}{6} & \frac{1}{6} \\ 0 & 0 & 0 & \frac{4}{6} & \frac{1}{6} & \frac{1}{6} \\ 0 & 0 & 0 & 0 & \frac{5}{6} & \frac{1}{6} \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix} X \begin{bmatrix} \frac{1}{6} & \frac{1}{6} & \frac{1}{6} & \frac{1}{6} & \frac{1}{6} & \frac{1}{6} \\ 0 & \frac{2}{6} & \frac{1}{6} & \frac{1}{6} & \frac{1}{6} & \frac{1}{6} \\ 0 & 0 & \frac{3}{6} & \frac{1}{6} & \frac{1}{6} & \frac{1}{6} \\ 0 & 0 & 0 & \frac{4}{6} & \frac{1}{6} & \frac{1}{6} \\ 0 & 0 & 0 & 0 & \frac{5}{6} & \frac{1}{6} \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

$$P^{(2)} = \frac{1}{36} \begin{bmatrix} 1 & 3 & 5 & 7 & 9 & 11 \\ 0 & 4 & 5 & 7 & 9 & 11 \\ 0 & 0 & 9 & 7 & 9 & 11 \\ 0 & 0 & 0 & 16 & 9 & 11 \\ 0 & 0 & 0 & 0 & 25 & 11 \end{bmatrix}$$

$$P^{(3)} = P^{(2)} X P = \frac{1}{36}\begin{bmatrix} 1 & 3 & 5 & 7 & 9 & 11 \\ 0 & 4 & 5 & 7 & 9 & 11 \\ 0 & 0 & 9 & 7 & 9 & 11 \\ 0 & 0 & 0 & 16 & 9 & 11 \\ 0 & 0 & 0 & 0 & 25 & 11 \\ 0 & 0 & 0 & 0 & 0 & 36 \end{bmatrix} X \begin{bmatrix} \frac{1}{6} & \frac{1}{6} & \frac{1}{6} & \frac{1}{6} & \frac{1}{6} & \frac{1}{6} \\ 0 & \frac{2}{6} & \frac{1}{6} & \frac{1}{6} & \frac{1}{6} & \frac{1}{6} \\ 0 & 0 & \frac{3}{6} & \frac{1}{6} & \frac{1}{6} & \frac{1}{6} \\ 0 & 0 & 0 & \frac{4}{6} & \frac{1}{6} & \frac{1}{6} \\ 0 & 0 & 0 & 0 & \frac{5}{6} & \frac{1}{6} \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

Problem - 06: A man's smoking habits are as follows. If he smokes filter cigarettes one week, he switches to non filter cigarettes the next week with probability 0.2. On the other hand, if he smokes non filter cigarettes one week there is a probability of 0.7 that he will smoke non filter cigarettes the next week as well. In the long run how often does he smokes filter cigarettes?

Solution: Here, there are two possible states, $A =$ represents the smoking filter cigarettes and $B =$ represents the smoking non filter cigarettes. The state space $= \begin{bmatrix} A & B \end{bmatrix}$. The corresponding transition probability matrix P is

$$P = \begin{bmatrix} 0.8 & 0.2 \\ 0.3 & 0.7 \end{bmatrix} = \begin{bmatrix} \frac{8}{10} & \frac{2}{10} \\ \frac{3}{10} & \frac{7}{10} \end{bmatrix} = \frac{1}{10} \begin{bmatrix} 8 & 2 \\ 3 & 7 \end{bmatrix}$$

In long run we have to find the unique fixed probability vector $v = \begin{bmatrix} x & y \end{bmatrix}$ such that $vP = v$

$$\begin{bmatrix} x & y \end{bmatrix} X \frac{1}{10} \begin{bmatrix} 8 & 2 \\ 3 & 7 \end{bmatrix} = \begin{bmatrix} x & y \end{bmatrix}$$

$$\begin{bmatrix} 8x+3y & 2x+7y \end{bmatrix} = 10 \begin{bmatrix} x & y \end{bmatrix}$$

$$\begin{bmatrix} 8x+3y & 2x+7y \end{bmatrix} = \begin{bmatrix} 10x & 10y \end{bmatrix}$$

Equating on both sides, we get $8x+3y = 10x$ and $2x+7y = 10y$, by solving these equations $x = \frac{3}{5}$ and $\frac{2}{5}$

Hence $v = \begin{bmatrix} x & y \end{bmatrix} = \begin{bmatrix} \frac{3}{5} & \frac{2}{5} \end{bmatrix} = \begin{bmatrix} P_A & P_B \end{bmatrix}$ Therefore, in the long run, he will smoke filter cigarettes $\frac{3}{5}$ or 60% of the time.

Problem - 07: A player has Rs.300. At each play of a game, he loses Rs.100 with probability $\frac{3}{4}$ but wins Rs.200 with probability $\frac{1}{4}$. He stops playing if he has lost his Rs.300 or he has won at least Rs.300. (a) Determine the transition probability matrix of the Markov chain. Find the probability that there are at least 4 plays to the game.

Solution: For a given problem, there are seven possible states, which are denoted by $a_0, a_1, a_2, a_3, a_4, a_5, a_6$. The state that he has Rs.i, hundreds, that is, the state a_0 represents he has Rs. 0, the state a_1 represents he has Rs. 100, the state a_2 represents he has Rs. 200, the state a_3 represents he has Rs. 300, the state a_4 represents he has Rs. 400, the state a_5 represents he has Rs. 500 and the state a_6 represents he has Rs. 600.

The state space $= \begin{bmatrix} a_0, & a_1, & a_2, & a_3, & a_4, & a_5, & a_6 \end{bmatrix}$

(a) The corresponding transition probability matrix P is given by

$$P = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{3}{4} & 0 & 0 & \frac{1}{4} & 0 & 0 & 0 \\ 0 & \frac{3}{4} & 0 & 0 & \frac{1}{4} & 0 & 0 \\ 0 & 0 & \frac{3}{4} & 0 & 0 & \frac{1}{4} & 0 \\ 0 & 0 & 0 & \frac{3}{4} & 0 & 0 & \frac{1}{4} \\ 0 & 0 & 0 & 0 & \frac{3}{4} & 0 & \frac{1}{4} \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

(b) The initial probability vector that there are at least 4 plays to the game is

$$P^{(0)} = \begin{bmatrix} 0, & 0, & 0, & 1, & 0, & 0, & 0 \end{bmatrix}$$

After the first play, The probabilities are

$$P^{(1)} = P^{(0)} X P = \begin{bmatrix} 0, & 0, & \frac{3}{4}, & 0, & 0, & \frac{1}{4}, & 0 \end{bmatrix}$$

After the second play, The probabilities are

$$P^{(2)} = P^{(1)}XP = \begin{bmatrix} 0, & \frac{9}{16}, & 0, & 0, & \frac{6}{16}, & 0, & \frac{1}{16} \end{bmatrix}$$

After the third play, The probabilities are

$$P^{(3)} = P^{(2)}XP = \begin{bmatrix} \frac{27}{64}, & 0, & 0, & \frac{27}{64}, & 0, & 0, & \frac{10}{64} \end{bmatrix}$$

After the fourth play, The probabilities are

$$P^{(4)} = P^{(3)}XP = \begin{bmatrix} \frac{27}{64}, & 0, & \frac{81}{256}, & 0, & 0, & \frac{27}{256}, & \frac{10}{64} \end{bmatrix}$$

Problem - 08: Every year, a man trades his car for a new car. If he has a Maruti, he trades it for an Ambassador. If he has an Ambassador, he trades it for a Santro. However, if he has a Santro, he is just as likely to trade it for a new Santro as to trade it for a Maruti or an Ambassador. In 2000 he bought his first car, which was a Santro. (i) Find the probability that he has (a) 2002 Santro (b) 2002 Maruti (c) 2003 Ambassador (d) 2003 Santro (ii) In the long run, how often will he have a Santro ?

Solution: (i) Here, there are three possible states, a_1, a_2 and a_3. The state a_1 represents he trades for a Maruti car, the state a_2 represents he trades for a Ambassador car and the state a_3 represents he trades for a Santro car then the state space $= \begin{bmatrix} a_1, & a_2, & a_3 \end{bmatrix}$

. The transition probability matrix P is given by

$$P = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ \frac{1}{3} & \frac{1}{3} & \frac{1}{3} \end{bmatrix}$$

$P_{11} = P\Big(a_1/a_1\Big) = 0 =$ From the state a_1 to the state a_1 =The probability that he has a Maruti car, given that he trades the Maruti car.
$P_{12} = P\Big(a_1/a_2\Big) = 1 =$ From the state a_1 to the state $a_2 =$ The probability that he has a Maruti car, given that he trades the Ambassador car.
$P_{13} = P\Big(a_1/a_3\Big) = 0 =$ From the state a_1 to the state $a_3 =$ The probability that he has a Maruti car, given that he trades the Santro car.

$P_{21} = P\left(a_2/a_1\right) = 0 =$ From the state a_2 to the state $a_1 =$ The probability that he has a Ambassador car, given that he trades the Maruti car.
$P_{22} = P\left(a_2/a_2\right) = 0 =$ From the state a_2 to the state $a_2 =$ The probability that he has a Ambassador car, given that he trades the Ambassador car.
$P_{23} = P\left(a_2/a_3\right) = 1 =$ From the state a_2 to the state $a_3 =$ The probability that he has a Ambassador car, given that he trades the Santro car.
$P_{31} = P\left(a_3/a_1\right) = \frac{1}{3} =$ From the state a_3 to the state $a_1 =$ The probability that he has a Santo car, given that he trades the Maruti car.
$P_{32} = P\left(a_3/a_2\right) = \frac{1}{3} =$ From the state a_3 to the state $a_2 =$ The probability that he has a Santo car, given that he trades the Ambassador car.
$P_{33} = P\left(a_3/a_3\right) = \frac{1}{3} =$ From the state a_3 to the state $a_3 =$ The probability that he has a Santo car, given that he trades the Santro car.

In the year 2000, he has a Santo car then the initial probability vector is $P^{(0)=\left(0,0,1\right)}$
(a) To reach 2002. we need to compute $P^{(2)} = P^{(0)} X P^2$

$$P^2 = P.P = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ \frac{1}{3} & \frac{1}{3} & \frac{1}{3} \end{bmatrix} X \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ \frac{1}{3} & \frac{1}{3} & \frac{1}{3} \end{bmatrix} = \begin{bmatrix} 0 & 0 & 1 \\ \frac{1}{3} & \frac{1}{3} & \frac{1}{3} \\ \frac{1}{9} & \frac{4}{9} & \frac{4}{9} \end{bmatrix}$$

Now, compute

$$P^{(2)} = P^{(0)} X P^2$$

$$P^{(2)} = P^{(0)} X P^2 = \begin{bmatrix} 0, & 0, & 1 \end{bmatrix} X \begin{bmatrix} 0 & 0 & 1 \\ \frac{1}{3} & \frac{1}{3} & \frac{1}{3} \\ \frac{1}{9} & \frac{4}{9} & \frac{4}{9} \end{bmatrix} = \begin{bmatrix} \frac{1}{9}, & \frac{4}{9}, & \frac{4}{9} \end{bmatrix}$$

$$P^{(2)} = \begin{bmatrix} P_{a_1}^{(2)}, & P_{a_2}^{(2)}, & P_{a_3}^{(2)} \end{bmatrix} = \begin{bmatrix} \frac{1}{9}, & \frac{4}{9}, & \frac{4}{9} \end{bmatrix}$$

(a) The probability that, he has a Santro car in the year 2002 is $P_{a3}^{(2)} = \frac{4}{9}$.
(b) The probability that, he has a Maruti car in the year 2002 is $P_{a1}^{(2)} = \frac{1}{9}$.
(c) The probability that, he has a Ambassador car in the year 2003, we need to compute the $P_{a_2}^{(3)}$.

$$P^{(3)} = P^{(0)} X P^3$$

Similarly simplify and find the required probabilities

(ii) In the long run, how often will he have a Santro, we have to find the unique fixed probability vector $v = \begin{bmatrix} x & y & Z \end{bmatrix}$ such that $vP = v$

$$\begin{bmatrix} x & y & z \end{bmatrix} X \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ \frac{1}{3} & \frac{1}{3} & \frac{1}{3} \end{bmatrix} = \begin{bmatrix} x & y & z \end{bmatrix}$$

II - Queuing Theory:

Introduction:

What is a queue: A queue is a waiting line and queue's are a part of every day life.

Examples:

1. Queue's to buy a movie tickets.
2. Queue's to make a bank deposits.
3. Aeroplane's are waiting to take off or landing.

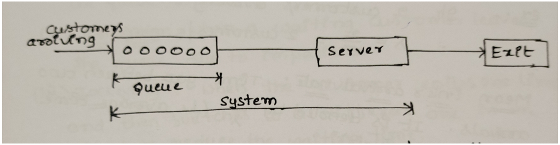
Objective of the Queueing Theory: The main objective or goal of a queueing theory is to form an analytical or mathematical models of the customers needing services and use that models to predict queue lengths and waiting times.

Use of queueing models: The queueing models are very useful to determining how to operate a queueing system in the most effective way and these models can be used to answer the questions like the following.

1. What is the expected number of customers present in the queue.
2. What is the expected time that a customers spends in the queue.
3. What fraction of the time each server is idle.
4. What is the probability distribution of the number of customers present in the queue.
5. What is the probability distribution of a customers waiting time.
6. How many servers should be employed, so that only 1% of all customers will have to wait a particular time etc.

Why Queue is formed?

Queue is formed if the service required by the customer is not available immediately. If the customers(people) arrive frequently they should wait to get their turn then the only way is to form a queue called the waiting line. Here the people(customer) arriving are called **Customers** and the person attending the customer is called a **server**.



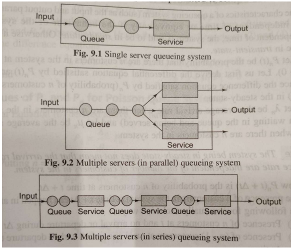


Figure 2: Queue Representation

In any queuing system, the aim is to design the system in such a way that the average(mean) waiting time of the customer is minimized and utilization of the server is managed above the desired level.

Elements of Queueing System:
A queueing system consists of the following elements

1. The arrival pattern.
2. The service pattern.
3. The queue discipline.
4. Customer behavior.

(i) - Arrival Pattern: The arrival pattern tells the way in which the customers arrive and join the system. Generally the arrival pattern is random. To study the randomness of arrival pattern we use probability distributions. In general, we use poisson distribution to study the arrival pattern.
Arrival Rate: The number of customers arriving per unit of time is called as arrival rate.
Mean arrival rate OR Average arrival rtae: The average rate at which customer arriving into the system. The average arrival rate is denoted by λ.
Example: If two customers arriving in every 5 minutes, then the average rate is $\lambda = \frac{2}{5} = (2 \text{ customers})/(5 \text{ minutes})$.
Mean inter arrival rate: The time gap between two arrivals is called as mean inter arrival rate and it is given by $\frac{1}{\lambda}$. (that is, the average time between two arrivals).

(ii) - Service Pattern: The service pattern tells how many customers can be served at a time.
Service rate: The number of customers served per unit of time is called as service rate.
Mean service rate: The mean service rate or The average service rate at which the customer is served in the system and it is represented by μ.
Example: If one customer served in two minutes in average then the average rate is $\mu = \frac{1}{2} =$ (one customer)/ (2 minutes).
Mean service time: The time gap between two services is called as mean serive time or average service time and it is given by $\frac{1}{\mu}$.
(iii) - Queue Discipline: The queue discipline is classified into the following four types

1. FIFO (OR) FCFS = First In First Out (OR) First Come First Served
2. LIFO = Last In First Out
3. SIRO = Service In Random Order (OR) Selection for service in Random Order.
4. PIR = Priority In Selection

(iv) - Customer Behavior: Generally the behavior of the customers is classified in to four types, they are

1. **Balking:** A customer may leave the queue because the queue is too long and he was no time to wait.
2. **Reneging:** When a waiting customer leaves the queue due to impatience.
3. **Jockeying:** When the customer enters one line and then switches to a different one in an effort to reduce the waiting time.
4. **Priority:** Some customers are served before others regardless of their order of arrival.

Kendall's Notation: The Kendall's notation (OR) representation of the queuing theory is **(a/b/c):(d/e)**
where
a = Inter - Arrival Distribution
b = Service time Distribution
c = Number of Servers (OR) Channel's
d = System capacity
e = Queue Discipline

The symbols $'a'$ and $'b'$ usually take one of the following
M = For Markovian (OR) Poisson distribution (OR) Exponential distribution.
G = For arbitrary (OR) General distribution
D = For fixed (OR) Deterministic Distribution
E_k = For k - stage Erlang Distribution
Classification of Queuing Models: The queuing models are classified into four models.

1. (M/M/1) : (∞/FIFO) - Single server with infinite capacity.
2. (M/M/1) : (k/FIFO) - Single server with finite capacity.
3. (M/M/S) : (∞ / FIFO) - Multiple server with infinite capacity.
4. (M/M/S) : (k/FIFO) - Multiple servers with finite capacity.

M/G/1 - Poisson Input/ General Out put / Single Server
Model - I (M/M/1) : (∞/FIFO) Single server with infinite capacity
Here, the mean arrival rate $(\lambda) <$ The mean service rate (μ)
Formulas for the model - I
Formula - 01: n = the number of customers in the system.
Formula - 02: $N(t)$ = The number of customers in the system at time t.
Formula - 03: The probability that the server is busy (OR) Traffic intensity (OR) Utilization factor $\rho = \frac{\lambda}{\mu} < 1$.
Formula - 04: The probability that the server is idle $= P_0 = 1 - \frac{\lambda}{\mu}$.

Formula - 05: The expected idle time = P_0 X Total service time.
Formula - 06: The probability that there are n customers in the system

$$P_n = \left(\frac{\lambda}{\mu}\right)^n P_0$$

Formula - 07: The probability that the number of customers is greater than a given number k is

$$P(N \geq k) = \left(\frac{\lambda}{\mu}\right)^k$$

(OR)

$$P(N > k) = \left(\frac{\lambda}{\mu}\right)^{k+1}$$

Formula - 08: The expected number of customers in the system (including the one who is being served) (OR) The average queue length is denoted by L_S and it is given by

$$L_S = \frac{\lambda}{\mu - \lambda}$$

Formula - 09: The expected number of customers in the queue (OR) The average queue length is dented by L_q and it is given by

$$L_q = \frac{\lambda^2}{\mu(\mu - \lambda)}$$

Formula - 10: The expected waiting time of a customer in the system is dented by W_S and it is given by
$W_S =$ (The expected number of customers in the system)/ (Arrival rate)

$$W_S = \frac{L_S}{\lambda} = \frac{\lambda}{\lambda(\mu - \lambda)} = \frac{1}{\mu - \lambda}$$

Hence,

$$W_S = \frac{1}{\mu - \lambda}$$

Formula - 11: The expected waiting time in the queue is dented by W_q and it is given by
$W_q = W_S -$ Time in service(The mean service time)

$$W_q = W_S - \frac{1}{\mu} = \frac{1}{\mu - \lambda} - \frac{1}{\mu} = \frac{\lambda}{\mu(\mu - \lambda)}$$

Therefore,

$$W_q = \frac{\lambda}{\mu(\mu - \lambda)}$$

Formula - 12: The expected length of non - empty queue is dented by L_n and it is given by

$$L_n = \frac{L_q}{P(n > 1)} = \frac{\frac{\lambda^2}{\mu(\mu-\lambda)}}{\left(\frac{\lambda}{\mu}\right)^2} = \frac{\mu}{(\mu - \lambda)}$$

Hence,

$$L_n = \frac{\mu}{\mu - \lambda}$$

Formula - 13: The probability that the queue is non - empty is given by

$$P(n > 1) = 1 - P_0 - P_1 = 1 - (1 - \frac{\lambda}{\mu}) - \frac{\lambda}{n}(1 - \frac{\lambda}{n})$$

$$P(n > 1) = \left(\frac{\lambda}{\mu}\right)^2$$

Model - I (M/M/1) : (∞/FIFO) Single Server with Infinite Capacity

Problem - 01: A supermarket has a single cashier. During peak hours, customers arrive at a rate of 20 per hour. The average number of customers that can be processed by the cashier is 24 per hour. Calculate
(a) The probability that the cashier is idle (b) The average number of customers in the queuing system (c) The average time a customer spends in the system (d) The average number of customers in the queue (e) The average time a customer spends in the queue waiting for service.

Solution: By data, The mean arrival rate = customer per time = $\lambda = 20$ per hour and the mean service rate = customer per time = $\mu = 24$ per hour.
(a) The probability that the cashier is idle

$$P_0 = 1 - \frac{\lambda}{\mu} = 1 - \frac{20}{24} = \frac{1}{6} = 0.16667$$

(b) The average number of customers in the queuing system

$$L_S = \frac{\lambda}{\mu - \lambda} = \frac{20}{24 - 20} = \frac{20}{4} = 5$$

(c) The average time a customer spends in the system

$$W_S = \frac{1}{\mu - \lambda} = \frac{1}{24 - 20} = \frac{1}{4} hour$$

(d) The average number of customers in the queue

$$L_q = \frac{\lambda^2}{\mu(\mu-\lambda)} = \frac{20^2}{24(24-20)} = 4.16667$$

(e) The average time a customer spends in the queue waiting for service

$$W_q = \frac{\lambda}{\mu(\mu-\lambda)} = \frac{20}{24(24-20)} = \frac{5}{24} hour.$$

Problem - 02: Customers arrive at a sales counter manned by a single person according to a Poisson process with a mean rate of 20 per hour. The time required to serve a customer has an exponential distribution with a mean of 100 seconds. Find the average waiting time of a customer.

Solution: Given, The mean arrival rate $= \lambda = 20$ per hour and the mean service rate $= \mu = \frac{1}{100}$ seconds $= \frac{60X60}{100} = 36$ per hour.

(a) The average waiting time of customer in the queue

$$W_q = \frac{\lambda}{\mu(\mu-\lambda)} = \frac{20}{36(36-20)} = \frac{5}{144} hour.$$

b) The average waiting time of customer in the system

$$W_S = \frac{1}{\mu-\lambda} = \frac{1}{36-20} = \frac{1}{16} hour$$

Problem - 03: Customers arrive at a watch repair shop according to a Poisson process at a rate of one per every 10 minutes and the service time is exponential with mean 8 minutes. Find the average number of customers in the shop, average waiting time a customer spends in the shop and the average time a customer spends in the queue for service.

Solution: By data, The mean arrival rate $= \lambda = \frac{1}{10}$ per minute $= \frac{60}{10}$ per hour $= 6$ per hour and the mean service rate $= \mu = \frac{1}{8}$ seconds $= \frac{60}{8} = 7.5$ per hour.

(i) The average number of customers in the shop (system) is

$$L_S = \frac{\lambda}{\mu - \lambda} = \frac{6}{7.5 - 6} = 4$$

(ii) The average waiting time of customer in the shop (system) is

$$W_S = \frac{1}{\mu - \lambda} = \frac{1}{7.5 - 6} = \frac{2}{3} hour$$

(iii) The average waiting time of customer in the queue is

$$W_q = \frac{\lambda}{\mu(\mu - \lambda)} = \frac{6}{7.5(7.5 - 6)} = \frac{8}{15} hour.$$

Problem - 04: In a city airport, flights arrive at a rate of 24 flights per day. It is known that the inter-arrival time follows an exponential distribution and the service time distribution is also exponential with an average of 30 minutes. Find the following
(a) The probability that the system will be idle (b) The mean queue size (c) The average number of flights in the queue (d) The probability that the size exceeds 7
Solution: By data, The mean arrival rate $= \lambda = 24$ per day $= \frac{24}{24}$ per hour $= 1$ per hour and the mean service rate $= \mu = \frac{1}{30}$ minute $= \frac{60}{30} = 2$ per hour.
(a) The probability that the cashier is idle is

$$P_0 = 1 - \frac{\lambda}{\mu} = 1 - \frac{1}{2} = \frac{1}{2} = 0.5$$

(b) The mean queue size i.e. The average number of flights in the system is

$$L_S = \frac{\lambda}{\mu - \lambda} = \frac{1}{2-1} = 1 flight$$

(c) The average number of flights in the queue

$$L_q = \frac{\lambda^2}{\mu(\mu - \lambda)} = \frac{1^2}{2(2-1)} = 0.5 flight$$

(d) The probability that the size exceeds $k = 7$ is

$$P(N > k) = \left(\frac{\lambda}{\mu}\right)^{k+1}$$

$$P(N > 7) = \left(\frac{1}{2}\right)^{8} = 0.0039$$

Problem - 05: Suppose that customers arrive at a Poisson rate of one per every 12 minutes and that the service time is exponential at a rate of one service per 8 minutes. What is

(a) The average number of customers in the system

(b) The average time a customer spends in the system

Solution: Given, The mean arrival rate $= \lambda = \frac{1}{12}$ per minute and the mean service rate $= \mu = \frac{1}{8}$ per minute.

(a) The average number of customers in the system

$$L_S = \frac{\lambda}{\mu - \lambda} = \frac{\frac{1}{12}}{\frac{1}{8} - \frac{1}{12}} = 2$$

(b) The average time a customer spends in the system

$$W_S = \frac{1}{\mu - \lambda} = \frac{1}{\frac{1}{8} - \frac{1}{12}} = 24 minute$$

Model - II (M/M/1) : (k/FIFO) - Single Server with Finite Capacity
The mean arrival rate = λ = one customer per time
the mean service rate = μ = one customer per time
Formulas for the Model - III (M/M/1) : (k/FIFO) - Single Server with Finite Capacity
Formula - 01: Traffic intensity $= \rho = \frac{\lambda}{\mu}$.
Formula - 02: The probability that an arriving customer will not wait is
P(The customer will not wait) $= P_0 = \frac{1-\rho}{1-\rho^{k+1}}$.
Formula - 03: The effective arrival rate is $\lambda_e = \mu(1 - P_0)$.
Formula - 04: The average number of customers in the system is

$$L_S = \frac{\rho}{1-\rho} - \frac{(k+1)\rho^{k+1}}{1-\rho^{k+1}}$$

Formula - 05: The expected time a customer spends in the system is

$$W_S = \frac{L_s}{\lambda_e}$$

Formula - 06: The average number of customers in the queue is

$$L_q = L_s - \frac{\lambda_e}{\mu}$$

Formula - 07: The expected waiting time of a customer in the queue is

$$W_q = \frac{L_s}{\lambda_e}$$

Problem - 06: Patients arrive at a doctor?s clinic according to Poisson distribution at a rate of 30 patients per hour. The waiting room does not accommodate more than 9 patients. Examination time per patient is exponential with mean rate of 20 per hour. Find the:
(a) Probability that an arriving patient will not wait (b) Effective arrival rate (c) Average number of patients in the clinic (d) Expected time a patient spends in the clinic (e) Average number of patients in the queue (f) Expected waiting time of a patient in the queue
Solution: Given, the mean arrival rate $= \lambda = 30$ per hour and the mean service rate $= \mu = 20$ per hour. Here $9 + 1 = 10$ (9 in the queue + 1 getting service). Therefore the Traffic intensity is

$$\rho = \frac{\lambda}{\mu} = \frac{30}{20} = \frac{3}{2} = 1.5$$

(a) The probability that an arriving patient will not wait = P(The patient will not wait) is

$$P_0 = \frac{1-\rho}{1-\rho^{(k+1)}} = \frac{1-1.5}{1-(1.5)^{11}} = 0.0058$$

(b) The effective arrival rate λ_e is

$$\lambda_e = \mu(1 - P_0) = 20(1 - 0.0058) = 19.8840$$

(c) The average number of patients in the clinic (system) is

$$L_S = \frac{\rho}{1-\rho} - \frac{(k+1)\rho^{k+1}}{1-\rho^{k+1}} = \frac{1.5}{1-1.5} - \frac{11(1.5)^{11}}{1-(1.5)^{11}} = 8.1287$$

(d) The expected time a patient spends in the clinic is

$$W_S = \frac{L_S}{\lambda_e} = \frac{8.1287}{19.8849} = 0.4088 per hour$$

(e) The average number of patients in the queue is

$$L_q = L_S - \frac{\lambda_e}{\mu} = 8.1287 - \frac{19.8840}{20} = 7.1345$$

(f) The expected waiting time of a patient in the queue is

$$W_q = \frac{L_q}{\lambda_e} = \frac{7.1345}{19.8849} = 0.3588 per hour$$

Problem - 07: A city has a one-person barber shop which can accommodate a maximum of 5 people at a time (4 waiting and 1 getting haircut). On average, customers arrive at the rate of 8 per hour and the barber takes 6 minutes for serving each customer. It is estimated that the arrival process is Poisson and the service time is an exponential random variable. Find:
(a) Percentage of time the barber is idle (b) Fraction of potential customers who will be turned away (c) Effective arrival rate of customers for the shop (d) Expected number of customers in the barber shop (e) Expected time a person spends in the barber shop (f) Expected number of customers waiting for a hair-cut (g) Expected waiting time of a customer in the queue
Solution: By data, the mean arrival rate $= \lambda = 8$ per hour and the mean service rate $= \mu = \frac{1}{6}$ per minute $= \frac{60}{6} = 10$ per hour. Here $4 + 1 = 5$ (4 in the queue + 1 getting service). Therefore the Traffic intensity is

$$\rho = \frac{\lambda}{\mu} = \frac{8}{10} = \frac{4}{5} = 0.8$$

(a) The probability that the time of the barber is idle is

$$P_0 = \frac{1-\rho}{1-\rho^{(k+1)}} = \frac{1-0.8}{1-(0.8)^6} = 0.2711$$

Therefore, the percentage of time the barber is idle is $= 0.2711X100 = 27.11\%$ (b) The fraction of potential customers who will be turned away is

$$P(K \geq 5) = \rho^k P_0 = (0.8)^5 X(0.2711) = 0.0888$$

(c) The effective arrival rate of customers for the shop is

$$\lambda_e = \mu(1 - P_0) = 10(1 - 0.2711) = 7.2890$$

d) The expected number of customers in the barber shop is

$$L_S = \frac{\rho}{1-\rho} - \frac{(k+1)\rho^{k+1}}{1-\rho^{k+1}} = \frac{0.8}{1-0.8} - \frac{6(0.8)^6}{1-(0.8)^6} = 1.8683$$

e) The expected time a person spends in the barber shop is

$$W_S = \frac{L_S}{\lambda_e} = \frac{1.8683}{7.2890} = 0.2563 perhour$$

f) The expected number of customers waiting for a hair-cut is

$$L_q = L_S - \frac{\lambda_e}{\mu} = 1.8683 - \frac{7.2890}{10} = 1.1394$$

g) The expected waiting time of a customer in the queue is

$$W_q = \frac{L_q}{\lambda_e} = \frac{1.1394}{7.2890} = 0.1563 perhour$$

Model - III (M/M/s) : (∞/FIFO) - Multiple Server with Infinite Capacity

The mean arrival rate = λ = one customer per time

the mean service rate = μ = one customer per time

The number of service counters = s

Formulas for the Model - III (M/M/s) : (∞/FIFO) - Multiple Server with Infinite Capacity

Formula - 01: The Probability that the system is idle is

$$P_0 = \left[\sum_{n=0}^{s-1} \frac{1}{(n)!} \left(\frac{\lambda}{\mu}\right)^n + \frac{1}{s!} \left(\frac{\lambda}{\mu}\right)^s \left(\frac{s\mu}{s\mu - \lambda}\right) \right]^{-1}$$

Formula - 02: The expected number of customers in the queue is

$$L_q = \left[\frac{1}{(s-1)!} \left(\frac{\lambda}{\mu}\right)^s \left(\frac{\lambda\mu}{(s\mu - \lambda)^2}\right) \right] X P_0$$

Formula - 03: The expected number of customers in the system is

$$L_s = L_q + \frac{\lambda}{\mu}$$

Formula - 04: The expected waiting time for a customer in the queue is

$$W_q = \frac{L_q}{\lambda}$$

Formula - 05: The expected time for a customer spends in the system is

$$W_s = W_q + \frac{1}{\mu}$$

Formula - 06: The probability that for a customer must wait before he gets service = The probability that the server are busy is

$$P(n \geq s) = \frac{1}{s!}\left(\frac{\lambda}{\mu}\right)^s \left(\frac{s\mu}{s\mu - \lambda}\right) X P_0$$

Formula - 07: The Traffic intensity of the system is

$$\rho = \frac{\lambda}{s\mu}$$

Problem - 08 A travel centre has three service counters to receive people who visit to book air tickets. The customers arrive in a Poisson distribution with the average arrival of 100 persons in a 10 hour service day. It has been estimated that the service time follows an exponential distribution. The average service time is 15 minutes. Find the

a) Expected number of customers in the queue b) Expected number of customers in the system c) Expected time a customer spends in the system d) Expected waiting time for a customer in the queue e) Probability that a customer must wait before he gets active

Solution: Given, The number of service counters $= s = 3$, the mean arrival rate $= \lambda = \frac{100}{10}$ per hour $= 10$ per hour, the mean service rate $= \mu = \frac{1}{15}$ per minutes $= \frac{60}{15} = 4$ and $\frac{\lambda}{\mu} = \frac{10}{4} = 2.5$

The Probability that the system is idle is

$$P_0 = \Big[\sum_{n=0}^{s-1} \frac{1}{(n)!}\Big(\frac{\lambda}{\mu}\Big)^n + \frac{1}{s!}\Big(\frac{\lambda}{\mu}\Big)^s\Big(\frac{s\mu}{s\mu-\lambda}\Big)\Big]^{-1}$$

$$P_0 = \Big[\sum_{n=0}^{2} \frac{1}{(n)!}\Big(2.5\Big)^n + \frac{1}{3!}\Big(2.5\Big)^3\Big(\frac{3X4}{3X4-10}\Big)\Big]^{-1}$$

Simplify the first term by Expanding the summation by substituting $n = 0, 1, 2$

$$P_0 = \left[\left\{\frac{1}{(0)!}\Big(2.5\Big)^0 + \frac{1}{(1)!}\Big(2.5\Big)^1 + \frac{1}{(2)!}\Big(2.5\Big)^2\right\} + \frac{1}{3!}\Big(2.5\Big)^3\Big(\frac{3X4}{3X4-10}\Big)\right]^{-1} = 0.0449$$

(a) The expected number of customers in the queue is

$$L_q = \left[\frac{1}{(s-1)!}\Big(\frac{\lambda}{\mu}\Big)^s\Big(\frac{\lambda\mu}{(s\mu-\lambda)^2}\Big)\right]XP_0$$

$$L_q = \left[\frac{1}{(2)!}\Big(2.5\Big)^3\Big(\frac{10X4}{(3X4-10)^2}\Big)\right]X(0.0449) = 3.5078$$

(b) The expected number of customers in the system is

$$L_s = L_q + \frac{\lambda}{\mu} = 3.5078 + 2.5 = 6.0078$$

(c) The expected time a customer spends in the queue is

$$W_q = \frac{L_q}{\lambda} = \frac{3.5078}{10} = 0.3508$$

(d) The expected waiting time for a customer in the system is

$$W_s = W_q + \frac{1}{\mu} = 0.3508 + \frac{1}{4} = 0.6008$$

(e) The Probability that a customer must wait before he gets service = The probability that the servers are busy is

$$P(n \geq s) = \frac{1}{s!}\left(\frac{\lambda}{\mu}\right)^s\left(\frac{s\mu}{s\mu - \lambda}\right)XP_0$$

$$P(n \geq 3) = \frac{1}{3!}\left(2.5\right)^3\left(\frac{3X4}{34 - 10}\right)X0.0449 = 0.7016$$

Problem - 09 A local hospital has 3 doctors for treating patients. The patients arrive at the hospital according to Poisson distribution with a an average of 8 per hour. On an average, a doctor takes about 15 minutes to treat each patient and the actual time taken is known to vary approximately exponentially around this average. Find

a) Traffic intensity of the system b) Probability that a customer has to wait for service c) Average number of customers waiting in the queue d) Average number of customers in the system e) Average waiting time a customer spends in the queue f) Expected time a customer spends in the system **Solution:** By data, The number of servers $= s = 3$, the mean arrival rate $= \lambda = 8$ per hour and the mean service rate $=$ $\mu = \frac{1}{15}$ per minutes $= \frac{60}{15} = 4$ per hour and $\frac{\lambda}{\mu} = 2$

(a) The traffic intensity of the system is

$$\rho = \frac{\lambda}{s\mu} = \frac{8}{3X4} = \frac{2}{3} = 0.6667$$

The Probability that the system is idle is

$$P_0 = \Big[\sum_{n=0}^{s-1} \frac{1}{(n)!}\Big(\frac{\lambda}{\mu}\Big)^n + \frac{1}{s!}\Big(\frac{\lambda}{\mu}\Big)^s\Big(\frac{s\mu}{s\mu-\lambda}\Big)\Big]^{-1}$$

$$P_0 = \Big[\sum_{n=0}^{2} \frac{1}{(n)!}\Big(2\Big)^n + \frac{1}{3!}\Big(2\Big)^3\Big(\frac{3X4}{3X4-8}\Big)\Big]^{-1}$$

Simplify the first term by Expanding the summation by substituting $n = 0, 1, 2$

$$P_0 = \Big[\Big\{\frac{1}{(0)!}\Big(2\Big)^0 + \frac{1}{(1)!}\Big(2\Big)^1 + \frac{1}{(2)!}\Big(2\Big)^2\Big\} + \frac{1}{3!}\Big(2\Big)^3\Big(\frac{12}{12-8}\Big)\Big]^{-1} = \frac{1}{9} = 0.1111$$

(b) The probability that a customer has to wait for service

$$P(n \geq s) = \frac{1}{s!}\Big(\frac{\lambda}{\mu}\Big)^s\Big(\frac{s\mu}{s\mu-\lambda}\Big)XP_0$$

$$P(n \geq 3) = \frac{1}{3!}\Big(2\Big)^3\Big(\frac{3X4}{3X4-8}\Big)X0.1111 = \frac{4}{9} = 0.4445$$

(c) The average number of customers waiting in the queue is

$$L_q = \Big[\frac{1}{(s-1)!}\Big(\frac{\lambda}{\mu}\Big)^s\Big(\frac{\lambda\mu}{(s\mu-\lambda)^2}\Big)\Big]XP_0$$

$$L_q = \left[\frac{1}{(2)!}\left(2\right)^3\left(\frac{8X4}{(3X4-8)^2}\right)\right]X(0.1111) = \frac{8}{9} = 0.8889$$

(d) The average number of customers in the system is

$$L_s = L_q + \frac{\lambda}{\mu} = 0.8889 + 2 = 2.8889$$

(e) The average waiting time a customer spends in the queue is

$$W_q = \frac{L_q}{\lambda} = \frac{0.8889}{8} = \frac{1}{9} = 0.1112$$

(f) The expected time a customer spends in the system is

$$W_s = W_q + \frac{1}{\mu} = \frac{1}{9} + \frac{1}{4} = 0.3611$$

Problem - 10 A petrol pump has 4 pumps. The service time follows an exponential distribution with a mean of 6 minutes and cars arrive for service in a Poisson process at the rate of 30 cars per hour. Find the average waiting time in the queue, average time spent in the system and the average number of cars in the system.

Solution: By data, The number of servers $= s = 4$, the mean arrival rate $= \lambda = 30$ per hour and the mean service rate $= \mu = \frac{1}{6}$ per minutes $= \frac{60}{6}$ per hour $= 10$ per hour. $\frac{\lambda}{\mu} = \frac{30}{10} = 3$.

(a) The Probability that the system is idle is

$$P_0 = \Big[\sum_{n=0}^{s-1} \frac{1}{(n)!}\Big(\frac{\lambda}{\mu}\Big)^n + \frac{1}{s!}\Big(\frac{\lambda}{\mu}\Big)^s\Big(\frac{s\mu}{s\mu-\lambda}\Big)\Big]^{-1}$$

$$P_0 = \Big[\sum_{n=0}^{2} \frac{1}{(n)!}\Big(3\Big)^n + \frac{1}{4!}\Big(3\Big)^4\Big(\frac{4X10}{4X10-30}\Big)\Big]^{-1}$$

Simplify the first term by Expanding the summation by substituting $n = 0, 1, 2$

$$P_0 = \Big[\Big\{\frac{1}{(0)!}\Big(3\Big)^0 + \frac{1}{(1)!}\Big(3\Big)^1 + \frac{1}{(2)!}\Big(3\Big)^2\Big\} + \frac{1}{4!}\Big(3\Big)^4\Big(\frac{40}{40-30}\Big)\Big]^{-1} = \frac{2}{53} = 0.0377$$

The average waiting time in the queue is

$$W_q = \frac{L_q}{\lambda}$$

The average number of customers waiting in the queue is

$$L_q = \Big[\frac{1}{(s-1)!}\Big(\frac{\lambda}{\mu}\Big)^s\Big(\frac{\lambda\mu}{(s\mu-\lambda)^2}\Big)\Big]XP_0$$

$$L_q = \Big[\frac{1}{(3)!}\Big(3\Big)^4\Big(\frac{30X10}{(4X10-30)^2}\Big)\Big]X(0.0377) = 1.5269$$

$$W_q = \frac{L_q}{\lambda} = \frac{1.5269}{30} == 0.0509$$

(b) The average time spent in the system is

$$W_s = W_q + \frac{1}{\mu} = 0.0509 + \frac{1}{10} = 0.1509$$

(c) The average number of cars in the system is

$$L_s = L_q + \frac{\lambda}{\mu} = 1.5269 + 3 = 4.5269$$

Problem - 11 A bank has three teller counters. The customers to the bank choose a teller counter at random. It is given that the arrivals to the teller counters are Poisson-distributed at the average rate ? and the mean service rate at the teller counters is ?/2 . Find the steady average queue at each counter.
Solution: By data, The number of servers = $s = 3$, the mean service rate = μ = (The mean arrival rate) /2 = $\frac{\lambda}{2}$ and $\frac{\lambda}{\mu} = 2$.
The steady average queue at each counter is

$$L_q = \left[\frac{1}{(s-1)!}\left(\frac{\lambda}{\mu}\right)^s \left(\frac{\lambda\mu}{(s\mu-\lambda)^2}\right)\right] X P_0$$

$$L_q = \Big[\frac{1}{s^2(s-1)!}\Big(\frac{\lambda}{\mu}\Big)^s\Big(\frac{\big(\frac{\lambda}{\mu}\big)}{(1-\frac{\lambda}{s\mu})^2}\Big)\Big]XP_0$$

The Probability that the system is idle is

$$P_0 = \Big[\sum_{n=0}^{s-1}\frac{1}{(n)!}\Big(\frac{\lambda}{\mu}\Big)^n + \frac{1}{s!}\Big(\frac{\lambda}{\mu}\Big)^s\Big(\frac{s\mu}{s\mu-\lambda}\Big)\Big]^{-1} = \Big[\sum_{n=0}^{s-1}\frac{1}{(n)!}\Big(\frac{\lambda}{\mu}\Big)^n + \frac{1}{s!}\Big(\frac{\lambda}{\mu}\Big)^s\Big(\frac{1}{1-\frac{\lambda}{s\mu}}\Big)\Big]^{-1}$$

$$P_0 = \Big[\sum_{n=0}^{s-1}\frac{1}{(n)!}\Big(2\Big)^n + \frac{1}{3!}\Big(2\Big)^3\Big(\frac{1}{1-\frac{2}{3}}\Big)\Big]^{-1}$$

Simplify the first term by Expanding the summation by substituting $n = 0, 1, 2$

$$P_0 = \Big[\Big\{\frac{1}{(0)!}\Big(2\Big)^0 + \frac{1}{(1)!}\Big(2\Big)^1 + \frac{1}{(2)!}\Big(2\Big)^2\Big\} + \frac{(2)^3}{3!}\Big(\frac{1}{1-\frac{2}{3}}\Big)\Big]^{-1} == \frac{1}{9} = 0.1112$$

$$L_q = \Big[\frac{1}{s^2(s-1)!}\Big(\frac{\lambda}{\mu}\Big)^s\Big(\frac{\big(\frac{\lambda}{\mu}\big)}{(1-\frac{\lambda}{s\mu})^2}\Big)\Big]XP_0$$

$$L_q = \Big[\frac{1}{3^2(2)!}\Big(2\Big)^3\Big(\frac{\big(2\big)}{(1-\frac{2}{3})^2}\Big)\Big]X\frac{1}{9} = 0.8889$$